

Chemical Reactions and Equations (Set B) — MCQ Worksheet

Science • Chapter 1 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Burning of LPG in air is an example of which type of reaction?

- (A) Combination and exothermic
- (B) Decomposition
- (C) Displacement
- (D) Endothermic only

Q2. When a piece of zinc is dropped in copper sulphate solution, the colour of solution:

- (A) Turns from blue to colourless
- (B) Turns from colourless to blue
- (C) Stays unchanged
- (D) Turns green

Q3. The substance that loses oxygen in a reaction is said to be:

- (A) Oxidised
- (B) Reduced
- (C) Neutralised
- (D) Hydrolysed

Q4. Which of these is a redox reaction?

- (A) $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$
- (B) $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
- (C) $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$
- (D) $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$

Q5. Reaction between barium chloride and sodium sulphate gives white precipitate of:

- (A) BaSO_4
- (B) Na_2SO_4
- (C) BaCl_2
- (D) NaCl

Q6. Iron + copper sulphate → iron sulphate + copper. This is a:

- (A) Combination
- (B) Decomposition
- (C) Displacement
- (D) Double displacement

Q7. Quicklime (CaO) + water → slaked lime is:

- (A) Endothermic
- (B) Exothermic
- (C) Photochemical
- (D) Reversible

Q8. The reaction $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$ (on heating with MnO_2) is:

- (A) Combination
- (B) Decomposition
- (C) Displacement
- (D) Double displacement

Q9. Why is hydrogen peroxide kept in dark coloured bottles?

- (A) To prevent photodecomposition into water and oxygen
- (B) Cost-saving
- (C) Looks better
- (D) It absorbs light

Q10. Reaction in which heat is absorbed from surroundings is called:

- (A) Exothermic
- (B) Endothermic
- (C) Combustion
- (D) Neutralisation

Q11. Symbol (g) in a chemical equation represents:

- (A) Solid
- (B) Liquid
- (C) Gas
- (D) Aqueous

Q12. When dilute HCl reacts with iron filings, the gas evolved is:

- (A) Oxygen
- (B) Hydrogen
- (C) Carbon dioxide
- (D) Chlorine

Q13. Brown fumes evolved on heating copper nitrate are of:

- (A) NO
- (B) NO_2
- (C) N_2O
- (D) N_2

Q14. Photosynthesis: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. The substance reduced is:

- (A) CO_2
- (B) H_2O
- (C) O_2
- (D) $\text{C}_6\text{H}_{12}\text{O}_6$

- Q15.** The change in colour of silver chloride from white to grey in sunlight is an example of:
 (A) Combination (B) Photochemical decomposition
 (C) Displacement (D) Double displacement
- Q16.** What is observed when iron nail is dipped in copper sulphate solution for some time?
 (A) Reddish-brown coat on nail; blue colour fades (B) Nail turns black; blue darkens
 (C) No change (D) Nail dissolves completely
- Q17.** Balanced equation: $\text{KNO}_3 \rightarrow \text{KNO}_2 + ?$
 (A) O (B) O_2
 (C) NO (D) N_2
- Q18.** Why are food articles often packed in nitrogen-filled foil?
 (A) To prevent oxidative rancidity (B) To increase shelf weight
 (C) For taste (D) To freeze them
- Q19.** An iron piece in hot water for many months forms reddish-brown coating because:
 (A) Iron reacts with oxygen and water to form hydrated iron(III) oxide (B) Iron reacts with water alone
 (C) Iron turns into copper (D) Water gets stale
- Q20.** When magnesium ribbon is burnt, gripping it with tongs is recommended because:
 (A) The flame is intensely bright and reaction releases heat (B) Ribbon is cold
 (C) Tongs are mandatory always (D) To prevent dust
- Q21.** Why does CO_2 turn limewater milky but turns colourless again on excess passage?
 (A) CaCO_3 forms first, then dissolves as soluble $\text{Ca}(\text{HCO}_3)_2$ (B) Reaction reverses
 (C) It evaporates (D) It crystallises
- Q22.** Balanced: $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$. This reaction is:
 (A) Combination + oxidation (B) Decomposition
 (C) Displacement (D) Neutralisation
- Q23.** Action of light on a film of silver bromide is used in:
 (A) Black-and-white photography (B) Cooking
 (C) Mirror making (D) Cleaning utensils
- Q24.** Why does the colour of copper sulphate change when an iron nail is dipped in it?
 (A) Iron displaces copper from copper sulphate (iron more reactive) (B) Iron dissolves
 (C) Copper sulphate evaporates (D) It does not change
- Q25.** A solution gives a brown precipitate with KI but no precipitate with AgNO_3 . The metal ion present is most likely:
 (A) Pb^{2+} (forms yellow PbI_2 ... actually gives yellow precipitate) (B) Cu^{2+} (forms brown CuI_2 precipitate)
 (C) Na^+ (D) K^+